```
> VerifyCutoffAndScalar := proc()
    local n, leftInteger, rightInteger, nextLeft, nextRight,
          s, wStar, product, k, term, ratio, psiLower, psiUpper,
          grid, lowerDisplay, upperDisplay;

    n := 60000;
    leftInteger := 300000000*n^5*1237^n;
    rightInteger := 1240^n;
    if 10^30*leftInteger >= rightInteger then
        error "the strict ratio bound for equation (39) fails";
    end if;
    printf("Equation (39) holds and its ratio is < 10^(-30)
PASS\n");

    nextLeft := 1237*(n+1)^5;
    nextRight := 1240*n^5;
    if nextLeft >= nextRight then
        error "the integer comparison for monotonicity fails";
    end if;
    printf("1237*60001^5 = %a < %a = 1240*60000^5   PASS\n",
           nextLeft, nextRight);

    s := 1546246693/5000000000;
    wStar := 23203769443/10^10;
    product := s*wStar;
    if s <= 0 or s >= 1 or wStar <= 1 or wStar >= 3 or product
>= 1 then
        error "the conditions on s and w_* fail";
    end if;
    printf("s*w_* = %a < 1   PASS\n", product);

    # The term of index k is (-w_*)^k*s^(k*(k-1)/2).
    # The ratio of successive magnitudes is w_* s^k <= w_* s < 1
    # for k >= 1. The sums through 13 and 12 enclose Psi(w_*,s).
    term, ratio, psiUpper := 1, wStar, 1;
    for k from 1 to 12 do
        term := -ratio*term;
        psiUpper := psiUpper+term;
        ratio := ratio*s;
    end do;
    psiLower := psiUpper-ratio*term;

    grid := 10^18;
    lowerDisplay := floor(grid*psiLower);
    upperDisplay := ceil(grid*psiUpper);
    if upperDisplay >= 0 then
```

```
        error "the alternating-series enclosure does not prove
  negativity";
    end if;
    printf("Psi(w_*,s) in [%a, %a] * 10^(-18)   PASS\n",
           lowerDisplay, upperDisplay);
    printf("The scalar inequalities imply T(s) < 3 by the
  argument in the paper.\n");
  end proc:

  VerifyCutoffAndScalar():
Equation (39) holds and its ratio is < 10^(-30)   PASS
1237*60001^5 = 96197136027196453237110123 7 < 964224000000000000000000000 = 1240
*60000^5   PASS
s*w_* = 3587875176637320199 9/50000000000000000000 < 1   PASS
Psi(w_*,s) in [-176228, -176227] * 10^(-18)   PASS
The scalar inequalities imply T(s) < 3 by the argument in the paper.
>
```